

FLAT: Smoothing the Rugged Landscape for Learnable, Sample-Efficient Traffic Calibration

Haopeng Deng

School of Future Transportation
Guangzhou Maritime University
Guangzhou 510725, China
qianyh@gmail.com

Shuo He*

Department of Civil and Environmental Engineering
Imperial College London
London SW7 2AZ, United Kingdom
sh4124@ic.ac.uk

Dayuan Wang

Department of Biostatistics
University of Florida
Gainesville, FL 32601, USA
dayuan.wang@ufl.edu

Abstract—Calibrating microscopic traffic models for digital twins is an expensive black-box optimization problem: tuning car-following and lane-changing parameters requires a full simulation run, affording only a tight budget per recalibration window. Matching raw trajectories yields a rugged objective that sparse surrogates cannot learn, reducing sequential acquisition to near-random probing. We present FLAT, which couples *what* to optimize with *where* to sample next. An eight-dimensional behavioral fingerprint smooths the parameter-error landscape, making the objective learnable from a few dozen samples; annealed lower-confidence-bound (LCB) acquisition then spends each remaining run where it most reduces error. The surrogate, interchangeable among a Gaussian process (GP), random forest (RF), or multi-layer-perceptron (MLP) ensemble, plugs into the same LCB loop. Across six heterogeneous real-world scenes, FLAT-GP achieves the lowest scene-averaged behavioral error, winning 6/6 scenes against SPSSA, GA, and CMA-ES and 5/6 against TPE under the matched budget. Some baselines need up to $4.4\times$ more simulations to match. Ablations show objective choice shifts final behavioral error by 81% on average, removing sequential LCB raises the six-scene mean by 20%, and surrogate choice shifts it by at most 4.2%, confirming gains trace to objective geometry and sequential allocation rather than surrogate capacity.

Index Terms—Surrogate-assisted optimization; traffic simulation calibration; behavioral fingerprint; sample-efficient learning.

I. INTRODUCTION

Operational decisions in urban traffic rely on near-real-time digital twins [1]. Their credibility rests on a microscopic simulator such as SUMO (Simulation of Urban MObility) [2] reproducing how vehicles move, not merely average flow; fidelity then depends on car-following [3] and lane-changing [4] parameters calibrated per location from trajectories. As twins roll on streaming observations, those parameters must be refreshed before lag undermines control. Each parameter setting is scored only by rerunning the full network, so calibration is an expensive, stochastic, black-box optimization problem with only a few minutes per recalibration window [5].

Two schools address this problem. The first searches the raw objective directly with general-purpose optimizers, including genetic algorithms (GA) [6], simultaneous perturbation stochastic approximation (SPSSA) [7], and the covariance matrix adaptation evolution strategy (CMA-ES) [8]. They are derivative-free

but sample-hungry; within the roughly hundred runs that fit such a window, even an evolution strategy completes only a handful of generations at best, retaining no memory of prior expensive evaluations. The second is surrogate-assisted, or Bayesian, optimization, which fits a cheap response-surface model to decide where to evaluate next. Jones *et al.* [9] formalized the idea, and the approach is now widely standard for costly simulators [10], including tree-structured Parzen estimation (TPE) [11], Gaussian-process (GP) [12], random-forest (RF) [13], and deep-learning multi-layer-perceptron (MLP) ensemble surrogates. The same paradigm now extends to large-scale traffic-model calibration [14]–[16]. Yet both schools treat the objective and the budget allocation as given, and no single optimizer wins consistently across setups [17].

That instability reflects a shared assumption: surrogate methods presuppose an objective learnable from few samples, yet in microscopic calibration the natural target, per-vehicle trajectory matching, is high-dimensional, rugged, and scene-dependent, preventing reliable surrogate learning and reducing acquisition to near-random probing. The literature keeps refining the optimizer or the surrogate family while leaving two upstream choices implicit: the *geometry of the objective* a surrogate must learn, and *how a fixed budget is split* between exploration and exploitation. The gap is therefore not accuracy but mechanism: without a learnable objective, sample-efficient acquisition has nothing to exploit.

We instantiate this as **FLAT** (**F**ingerprint-guided **L**earnable **A**cquisition for **T**raffic Calibration), which makes both choices explicit. For the objective, it replaces trajectory matching with a compact *behavioral fingerprint* that smooths the parameter-error landscape, enabling reliable surrogate learning from a few dozen runs. For the budget, an uncertainty-aware acquisition policy allocates the remaining simulation sequentially, annealing exploration into exploitation as the budget depletes. The framework is surrogate-agnostic, which lets us test whether efficiency comes from model capacity or from how the problem is posed. Fig. 1 overviews the framework.

Contributions.

- 1) We propose FLAT, a surrogate-acquisition framework that converts the rugged objective into a learnable surface via behavioral representation reshaping, enabling sample-efficient calibration within 10^2 evaluations per scene.

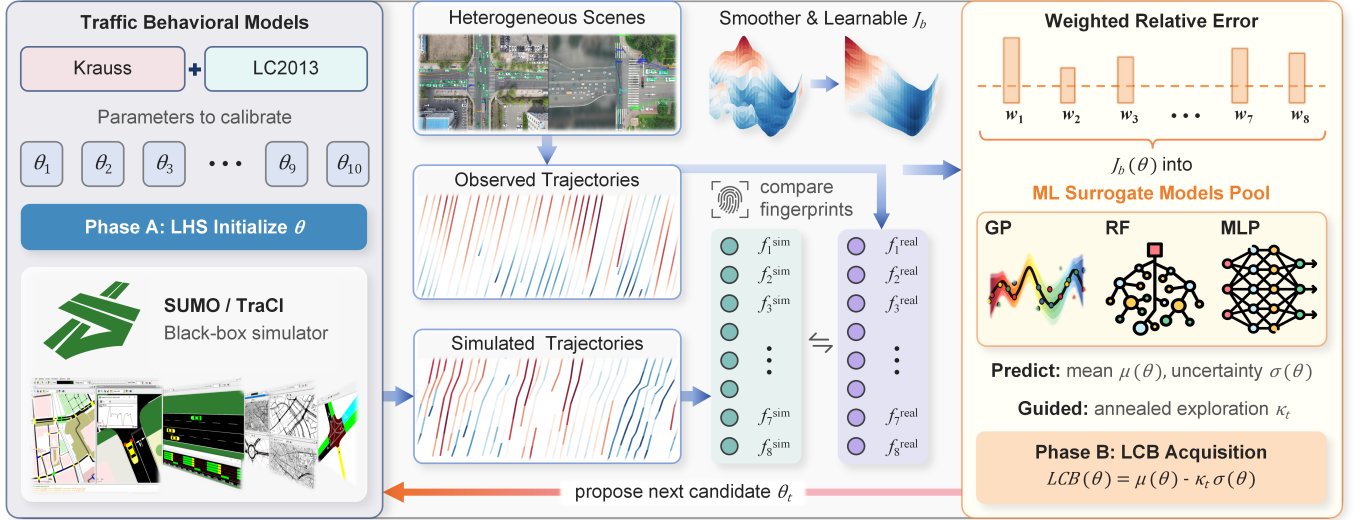


Fig. 1. FLAT overview. Black-box calibration of Krauss car-following and LC2013 lane-changing parameters in SUMO–TraCI on heterogeneous scenes. Behavioral fingerprints encode trajectories into a smoother parameter–error landscape, making $J_b(\theta)$ learnable for sample-efficient acquisition. Phase A explores θ by Latin-hypercube sampling (LHS); Phase B refits a GP, RF, or MLP surrogate and selects the next evaluation by annealed LCB.

- 2) We establish a consistent empirical advantage over four mechanistically diverse optimizer families across six real-world scenes, with some competitors requiring up to $4.4\times$ more simulations to reach equivalent quality.
- 3) We show through ablations that objective geometry and sequential allocation, not surrogate capacity, drive the gain; objective choice shifts error by 81%, sequential allocation by 20%, surrogate choice by at most 4.2%.

II. LITERATURE REVIEW

Punzo *et al.* [18] rank goodness-of-fit measures with Pareto-efficiency and indifference curves, exposing trade-offs in spacing, speed, and acceleration fit and yielding leader–follower guidelines; yet they stop at the pairwise-interaction scale and treat the optimizer as a black box. Schrader *et al.* [19] likewise score trajectory normalized NRMSE in SUMO-in-the-loop calibration, yet at full-network scale, which behavioral summary enters the objective remains unset when lane changing, queuing, and re-seeding stochastically reshape each evaluation; that choice must be fixed before optimization begins.

A parallel line matches distributional outputs rather than pointwise trajectories, shifting attention from paths to aggregate behavior. Abuamer and Tampère [20] show that Wasserstein distances on speed and headway improve parameter convergence consistency relative to RMSRE or Kolmogorov–Smirnov metrics under stochastic microsimulation; yet they optimize detector-level error with conventional SPSA or Bayesian backends and never ask whether the resulting loss is surrogate-fittable at online-twin budgets. High-dimensional Bayesian optimization exposes the same tension from the machine-learning side: Moriconi *et al.* [21] learn a low-dimensional feature space jointly with a Gaussian-process response surface, but fitting the embedding consumes evaluations that full-network microsimulation budgets rarely afford. Neither

improved metrics nor learned embeddings establish whether the loss is surrogate-fittable from a few dozen full-network runs, the regime that defines microscopic twin calibration.

The learning-for-optimization literature tackles objective geometry head-on. Zharmagambetov *et al.* [22] treat landscape surrogates as a design choice: construct a smooth decision loss rather than optimize through a rugged simulator-native loss; LANCER realizes this offline with neural mappings trained under many inner solver calls, far from refitting a response surface every few simulations. When the objective is instead held fixed and assumed learnable, surrogate-assisted optimization spends its budget on where to sample [10], via random-forest response surfaces [13], trust-region local search [23], or SUMO-backed benchmarks such as BO4Mob [16] for high-dimensional origin–destination estimation under aggregate sensor mismatch. No existing line jointly specifies a domain-informed, learnable objective for microscopic calibration and then expends a tight online budget through sequential surrogate refit; the setting remains distinct from offline landscape surrogates and from fixed-objective Bayesian optimization.

III. PROBLEM FORMULATION

Calibration tunes six Krauss car-following [3] and four LC2013 lane-changing [4] parameters in SUMO [2] via TraCI, the traffic control interface, as a ten-dimensional vector θ set at runtime; physical meanings in Appendix B. Trajectory matching scores spatiotemporal sequences *pointwise*:

$$J_{\text{traj}}(\theta) = \frac{1}{|\mathcal{V}|} \sum_{v \in \mathcal{V}} \frac{1}{T} \sum_{t=1}^T \|\mathbf{y}_v^{\text{sim}}(\theta, t) - \mathbf{y}_v^{\text{real}}(t)\|_2^2, \quad (1)$$

where $\mathbf{y}_v(t)$ denotes the simulated or observed state (e.g. position and speed) of vehicle $v \in \mathcal{V}$ at $t \in \{1, \dots, T\}$, and one seeks θ^* that minimizes J_{traj} .

Yet microscopic traffic is a stochastic, strongly coupled dynamical system, and minimizing J_{traj} is therefore ill-posed under a tight budget. Re-seeding reshuffles sim–real vehicle identity, while pointwise replay misses the stop–go structure, speed spreading, and queue dynamics that practical calibration requires. Relaxing to network-mean speed over time,

$$J_s(\theta) = \sqrt{\frac{1}{T} \sum_{t=1}^T (\bar{v}^{\text{sim}}(\theta, t) - \bar{v}^{\text{real}}(t))^2}, \quad (2)$$

still leaves a rugged, behaviorally misaligned landscape.

Instead, we compare ensemble distributional statistics. Each trajectory set $\mathcal{T}$ is mapped by a fixed $\phi : \mathcal{T} \rightarrow \mathbb{R}^K$, and we sum the per-component relative errors,

$$J_{\text{dist}}(\theta) = \sum_{k=1}^K \frac{|f_k^{\text{sim}}(\theta) - f_k^{\text{real}}|}{\max(|f_k^{\text{real}}|, \epsilon)}, \quad (3)$$

where f_k^{sim} and f_k^{real} are the k th components of ϕ for simulated and observed traffic. Because each simulation is expensive, and an online digital twin affords only a limited recalibration window, sample efficiency is the binding constraint.

IV. METHODOLOGY

FLAT’s principle: *spend each simulation where it matters most*. Two coupled decisions realize it—what to optimize and where to sample next—via a behavioral-fingerprint objective (§IV-A) and a sequential acquisition loop with a pluggable surrogate (§IV-B–§IV-C). Algorithm 1 details the steps.

Algorithm 1 FLAT calibration under simulation budget B .

Require: Scene config; real fingerprint $\mathbf{f}^{\text{real}}$; bounds; N_{init} ; schedule κ_t

Ensure: Calibrated θ^* and history $\{(\theta_i, J_b(\theta_i))\}_{i=1}^B$

- 1: Draw $\{\theta_i\}_{i=1}^{N_{\text{init}}}$ by LHS; evaluate J_b in SUMO ▷ Phase A
- 2: **for** $t = N_{\text{init}} + 1, \dots, B$ **do** ▷ Phase B
- 3: Fit surrogate (μ, σ) on $\{(\theta_i, J_b(\theta_i))\}_{i < t}$
- 4: $\theta_{\text{best}} \leftarrow \arg \min_{i < t} J_b(\theta_i)$; sample trust region around θ_{best}
+ global candidates; drop near-duplicates
- 5: $\theta_t \leftarrow \arg \min_{\theta} \mu(\theta) - \kappa_t \sigma(\theta)$
- 6: Evaluate $J_b(\theta_t)$ in SUMO; append
- 7: **end for**
- 8: **return** $\theta^* = \arg \min_i J_b(\theta_i)$

A. Behavioral Fingerprint Objective

Behavioral fingerprinting instantiates (3) at $K=8$ with ϕ drawn from core facility performance measures in the FHWA *Traffic Analysis Toolbox* [24] and the *Highway Capacity Manual* [25]. Domain-specified and fixed before search rather than learned from data, ϕ spends no evaluation budget on feature fitting. The fingerprint vector is

$$\mathbf{f} = (v_{\text{mean}}, v_{15}, v_{85}, \bar{a}^+, \bar{a}^-, a_5, a_{95}, s_{\text{stop}}), \quad (4)$$

comprising mean speed and 15th/85th percentiles, acceleration and deceleration at 5th/95th percentiles, and stop fraction.

Weights follow inverse-variance weighting (IVW), $w_k \propto 1/\text{Var}_k$ on per-vehicle references, normalized and held fixed; per-term clipping at 3 refines (3) into objective:

$$J_b(\theta) = \sum_{k=1}^8 w_k \min\left(\frac{|f_k^{\text{sim}}(\theta) - f_k^{\text{real}}|}{\max(|f_k^{\text{real}}|, \epsilon)}, 3\right). \quad (5)$$

Fig. 2 contrasts J_s and J_b on the XAM expressway; J_b smooths the parameter–error surface for surrogate guidance.

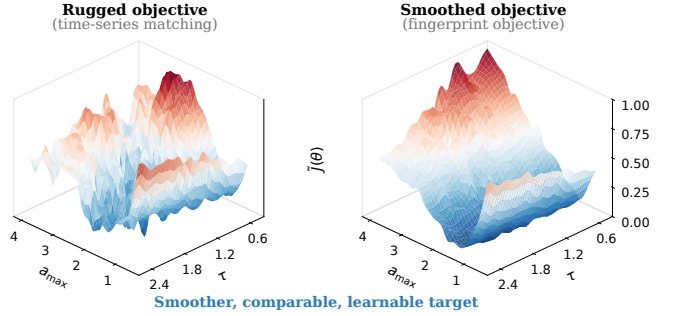


Fig. 2. Objective landscapes over maximum acceleration a_{max} and desired time headway τ on the XAM expressway, from 400 SUMO runs; surfaces normalized to $[0, 1]$. Left: J_s (2). Right: J_b (5).

B. Sequential Acquisition

The budget B is split between Phase A exploration and Phase B exploitation. Phase A draws N_{init} LHS [26] samples over θ in SUMO. Phase B spends the remaining $B - N_{\text{init}}$ simulations sequentially: at each step it (i) fits a surrogate returning $\mu(\theta)$ and $\sigma(\theta)$ on observed (θ, J_b) pairs, (ii) draws candidates from a trust region [23] around the incumbent plus a global pool, (iii) selects the candidate minimizing LCB

$$\text{LCB}(\theta) = \mu(\theta) - \kappa_t \sigma(\theta), \quad (6)$$

where κ_t is the annealed exploration–exploitation weight in (7), after discarding points too close to evaluated ones, and (iv) evaluates it in SUMO and appends the result.

The uncertainty term σ lets the acquisition explore where the surrogate is unsure and exploit where it is confident. Over the $B - N_{\text{init}}$ sequential steps t we anneal κ_t linearly,

$$\kappa_t = \kappa_0 + (\kappa_T - \kappa_0) \frac{t}{B - N_{\text{init}} - 1}, \quad \kappa_0 = 2.0, \kappa_T = 0.5, \quad (7)$$

shifting from exploration to exploitation as the budget depletes. On the same XAM slice (Fig. 3), Phase-B trials concentrate in the low- J_b valley after Phase-A coverage.

C. Pluggable ML Surrogates

Phase B treats every surrogate through the same interface: given training pairs $\mathcal{D} = \{(\theta_i, J_b(\theta_i))\}$, each refit returns $\mu(\theta)$ and $\sigma(\theta)$ for (6). RF and MLP aggregate M members via

$$\mu(\theta) = \frac{1}{M} \sum_{m=1}^M h_m(\theta), \quad \sigma(\theta) = \sqrt{\frac{1}{M} \sum_{m=1}^M (h_m(\theta) - \mu(\theta))^2}. \quad (8)$$

The GP instead reads μ and σ directly from the posterior ((10)).

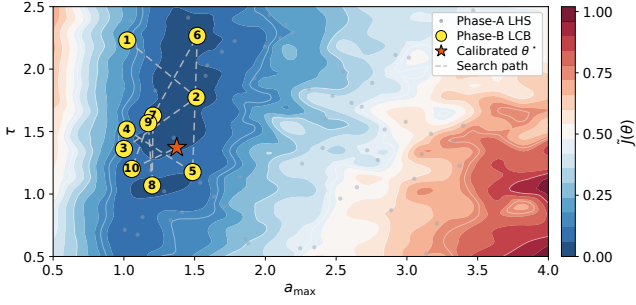


Fig. 3. Sequential acquisition in the $(a_{\max}, \tau)$ plane (400 SUMO evaluations, XAM expressway); $J(\theta)$ normalized to $[0, 1]$ for shape comparison. Grey: Phase-A LHS. Numbered: Phase-B LCB trials. Star: calibrated θ^* .

a) Gaussian Process: The GP [12] is a Bayesian non-parametric regressor with a Matérn-5/2 kernel ($\nu=\frac{5}{2}$), signal scale σ_f , length scales l_d , and noise σ_n :

$$k(\theta, \theta') = \sigma_f^2 \left(1 + \sqrt{5r} + \frac{5}{3}r \right) \exp(-\sqrt{5r}), \quad (9)$$

$$r = \sum_{d=1}^{10} (\theta_d - \theta'_d)^2 / l_d^2.$$

Inputs are standardized and J_b is z -scored before fitting; predictions are mapped back to the original scale. Given Gram matrix K and cross-covariance $\mathbf{k}_*$, the posterior at θ_* is

$$\mu(\theta_*) = \mathbf{k}_*^\top (K + \sigma_n^2 I)^{-1} \mathbf{y}, \quad (10)$$

$$\sigma^2(\theta_*) = k(\theta_*, \theta_*) - \mathbf{k}_*^\top (K + \sigma_n^2 I)^{-1} \mathbf{k}_*.$$

Hyperparameters maximize marginal likelihood at each refit.

b) Random Forest: The RF regressor [13] uses $M=500$ bootstrap trees with max. depth 20 on raw θ . Tree m returns the leaf-mean prediction

$$h_m^{\text{RF}}(\theta) = \sum_{\ell \in \mathcal{L}_m} c_{m\ell} \mathbf{1}\{\theta \in R_{m\ell}\}, \quad (11)$$

$$c_{m\ell} = \frac{1}{|\mathcal{I}_{m\ell}|} \sum_{i \in \mathcal{I}_{m\ell}} J_b(\theta_i),$$

where $\mathcal{L}_m$ is the leaf set of tree m , $R_{m\ell}$ are the leaf hyper-rectangles in Θ , and $\mathcal{I}_{m\ell}$ indexes training points in leaf ℓ . Inter-tree mean and spread supply μ and σ via (8).

c) MLP Ensemble: The higher-capacity MLP ensemble [27] uses $M=10$ bootstrap-trained two-layer (64, 64) member networks, each optimized with limited-memory Broyden-Fletcher-Goldfarb-Shanno (L-BFGS). Inputs are min-max scaled to the per-scene bounds,

$$\tilde{\theta} = (\theta - \theta_{\min}) / (\theta_{\max} - \theta_{\min}), \quad (12)$$

and member m predicts

$$h_m^{\text{MLP}}(\theta) = \bar{J} + s_J \hat{y}_m(\tilde{\theta}), \quad (13)$$

where $\hat{y}_m$ is the normalized network output and $\bar{J}$ and s_J are the mean and std. of J_b from the initial fit. Phase B warm-starts member weights rather than refitting from scratch. Inter-member mean and spread supply μ and σ via (8).

V. EXPERIMENTS

A. Setup

Datasets. We evaluate on six heterogeneous real-world scenes reconstructed as SUMO networks from SIND [28] and UTE [29] aerial trajectories (Fig. 4): signalized intersections (Tianjin, Changchun, Xi'an) spanning a low-to-high saturation range, and urban-expressway segments (YTDJ, RML, XAM) covering merge and weaving bottlenecks.

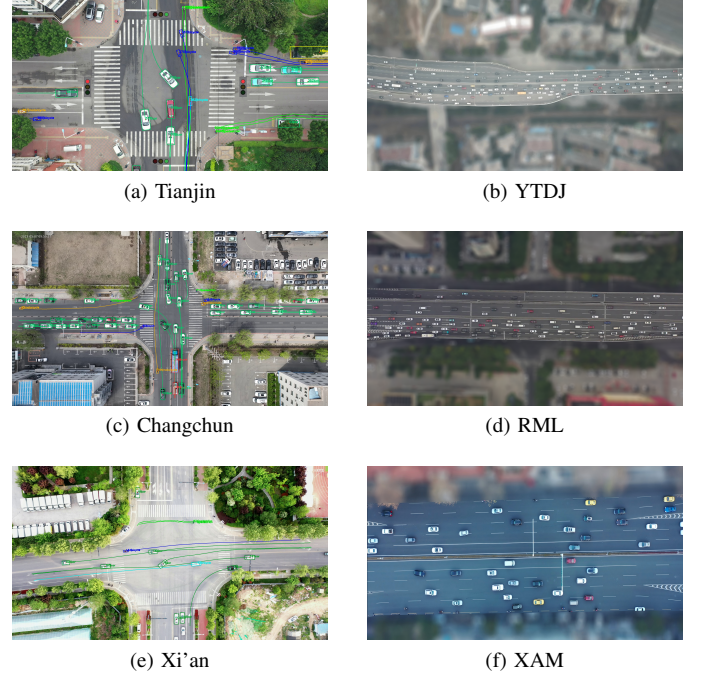


Fig. 4. Six heterogeneous evaluation scenes: *Left*: SIND signalized intersections. *Right*: UTE urban-expressway segments.

Budget. That recalibration window must finish within wall-clock $T \sim 10^2$ s [5]. Each serial SUMO-TraCI replay is second-scale; deployment overhead changes runtime but not the order of magnitude, so evaluations remain $\mathcal{O}(10^2)$ and $B=100$.

Protocol. All methods use identical parameter bounds and share $B=100$, objective J_b , IVW weights, and seeds $\{42, 101, 202, 303, 404\}$, with a fresh SUMO seed per run. FLAT uses $N_{\text{init}}=40$ and an 18% trust region; baseline settings are in Appendix C. We report best-so-far J_b .

Baselines. We instantiate FLAT with three interchangeable ML surrogates, GP, RF, and MLP, and compare against four optimizer families: SPSA [7] (gradient estimation), GA (population search), CMA-ES [8] (evolution strategy), and TPE [11] (density-estimation Bayesian optimizer).

Significance. We average the five runs within each scene and apply two-sided exact paired Wilcoxon tests [30] across the six scene-level observations. Statistical evidence is reported jointly with effect sizes and cross-scene win counts.

B. Best Under Equal Budget

Under equal budget, FLAT attains the lowest six-scene mean J_b (Table I). GP reaches 0.231, with RF and MLP at 0.241 and

0.240, well ahead of TPE (+11.3% vs. GP), GA (+19.5%), CMA-ES (+21.2%), and SPSA (+66.0%). Because the four baselines span distinct optimizer mechanisms, the margin is unlikely to reflect a coincidental match to one class.

TABLE I
SIX-SCENE MEAN J_b AT BUDGET 100 (MEAN $\pm$ SEED STD OVER FIVE SEEDS). $\Delta\%$: RELATIVE TO EACH FLAT VARIANT'S GLOBAL MEAN. WINS/6: FLAT-GP SCENE WINS; p : SCENE-LEVEL TEST.

Method	Mean $J_b \downarrow$	vs. GP	vs. RF	vs. MLP	Wins/6	p
FLAT-GP	0.231 ± 0.003	—	−4.0	−3.9	—	—
FLAT-RF	0.241 ± 0.009	+4.2	—	−0.1	—	—
FLAT-MLP	0.240 ± 0.006	+3.9	−0.1	—	—	—
TPE	0.257 ± 0.010	+11.3	+6.7	+6.9	5/6	0.0625
GA	0.276 ± 0.013	+19.5	+14.7	+14.8	6/6	0.0313
CMA-ES	0.280 ± 0.023	+21.2	+16.3	+16.4	6/6	0.0313
SPSA	0.383 ± 0.043	+66.0	+59.3	+59.5	6/6	0.0313

The scene-level results support a cross-scene advantage for FLAT (Table I). Fig. 5 shows FLAT still refining past the LHS-to-sequential transition while all four baselines stall, and the narrowest seed spread throughout (± 0.006 – 0.009 vs. 0.043 for SPSA). Fig. 6 cross-checks J_b against observed speed distributions (similarity $\bar{\rho}=0.75$), with expressways closest ($\rho \geq 0.80$, tight seed bands). Aligned v_{15} , $\bar{v}$, and v_{85} confirm J_b -weighted distributional terms.

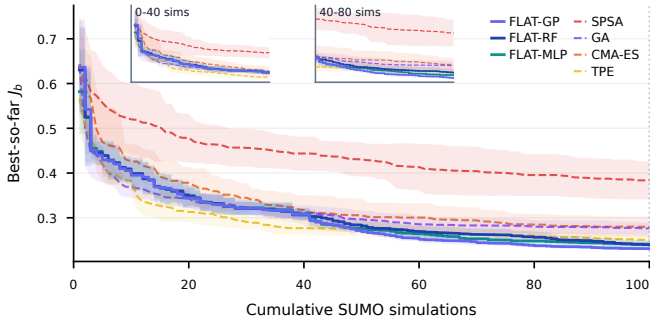


Fig. 5. Running best-so-far J_b for seven optimizers, averaged over six scenes and five seeds. Shaded bands: ± 1 std across seeds. Dotted line: budget cap $B=100$. Insets: zoomed 0–40 and 40–80 simulations.

C. Cheapest to Equal Quality

Equal-budget error (§V-B) does not establish whether baselines close the gap with more simulations. We extend each to $B=500$ and record, per scene $\times$ seed pair, the first run whose running-best J_b meets FLAT's per-scene mean at $B=100$ (Table II). Even TPE still requires $1.47\times$ – $2.21\times$ as many evaluations (± 139 – 162 simulations across pairs), while CMA-ES and GA average $2.76\times$ – $4.14\times$. Furthermore, only 28/30 pairs for TPE and 9/30 for SPSA reach that bar within the 500-sim cap; the gap is a failure to match, not just slower convergence. FLAT's sample efficiency offers clear practical value for real-world deployment under tight windows.

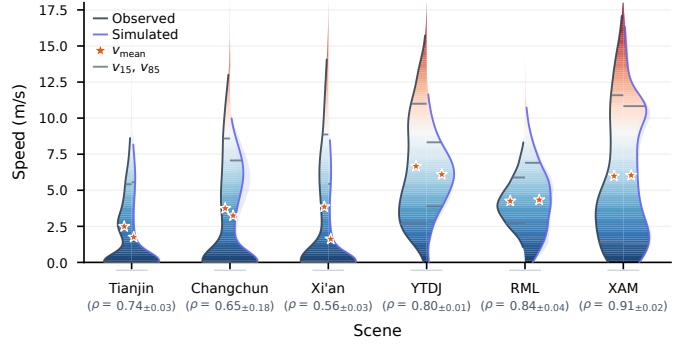


Fig. 6. Observed vs. FLAT-GP simulated speed distributions (five seeds); simulated mean kernel density estimation (KDE) ± 1 std envelope. Per-scene $\rho=1-D_{KS}$ (mean $\pm$ seed std); stars/ticks: v_{mean} , v_{15} , v_{85} .

TABLE II
SIMULATIONS NEEDED TO MATCH FLAT@100, MEAN $\pm$ STD OVER 30 SCENE $\times$ SEED PAIRS, BUDGET CAPPED AT 500. REACH: PAIRS REACHING THE QUALITY THRESHOLD WITHIN CAP, VS. FLAT-RF.

Baseline	vs. GP		vs. RF		vs. MLP		Reach $\uparrow$
	Sims $\downarrow$	Ratio	Sims $\downarrow$	Ratio	Sims $\downarrow$	Ratio	
TPE	221 ± 162	2.21 $\times$	167 ± 139	1.67 $\times$	147 ± 140	1.47 $\times$	28/30
CMA-ES	402 ± 108	4.02 $\times$	288 ± 148	2.88 $\times$	276 ± 108	2.76 $\times$	25/30
GA	414 ± 142	4.14 $\times$	360 ± 165	3.60 $\times$	305 ± 170	3.05 $\times$	15/30
SPSA	442 ± 131	4.42 $\times$	409 ± 159	4.09 $\times$	424 ± 154	4.24 $\times$	9/30

D. Sources of Performance Gain

We attribute the advantage to objective geometry, sequential acquisition, and surrogate capacity, and ablate each in turn.

Objective geometry dominates. Holding the optimizer and budget fixed, we train on J_s or J_b but always score θ^* by J_b . Training on J_s degrades FLAT-RF and TPE in 6/6 scenes (scene-level exact Wilcoxon $p=0.031$ each). Final J_b rises by 81% on average under FLAT-RF and more than doubles on Tianjin and YTDJ. Surrogates still fit J_s , yet the returned parameters miss behavioral alignment; the same degradation under TPE rules out an acquisition-specific artifact. J_b alone occupies the middle rung between (1) and (2) (Fig. 2). Appendix A traces this mechanism from landscape smoothness through learnability to calibrated uncertainty for (6).

TABLE III
OBJECTIVE ABLATION: FINAL J_b WHEN TRAINING ON J_b VS. J_s (ALWAYS SCORED BY J_b). FIVE-SEED MEAN $\pm$ STD; 6/6 SCENES, SCENE-LEVEL EXACT WILCOXON $p=0.031$ PER OPTIMIZER.

Scene	FLAT-RF (J_b)	FLAT-RF (J_s)	TPE (J_b)	TPE (J_s)
Tianjin	0.146 ± 0.011	0.345 ± 0.066	0.165 ± 0.010	0.327 ± 0.116
Changchun	0.168 ± 0.016	0.279 ± 0.041	0.167 ± 0.013	0.277 ± 0.068
Xi'an	0.430 ± 0.005	0.488 ± 0.014	0.426 ± 0.013	0.471 ± 0.017
YTDJ	0.223 ± 0.014	0.467 ± 0.129	0.284 ± 0.067	0.435 ± 0.102
RML	0.434 ± 0.022	0.514 ± 0.016	0.452 ± 0.008	0.620 ± 0.024
XAM	0.044 ± 0.008	0.105 ± 0.097	0.047 ± 0.008	0.234 ± 0.107

Sequential acquisition contributes. Holding J_b and $B=100$ fixed, we sweep N_{init} (Fig. 7); $N_{\text{init}}=100$ leaves no LCB step. The six-scene mean stays near 0.23 for $N_{\text{init}}=20$ –40 but rises

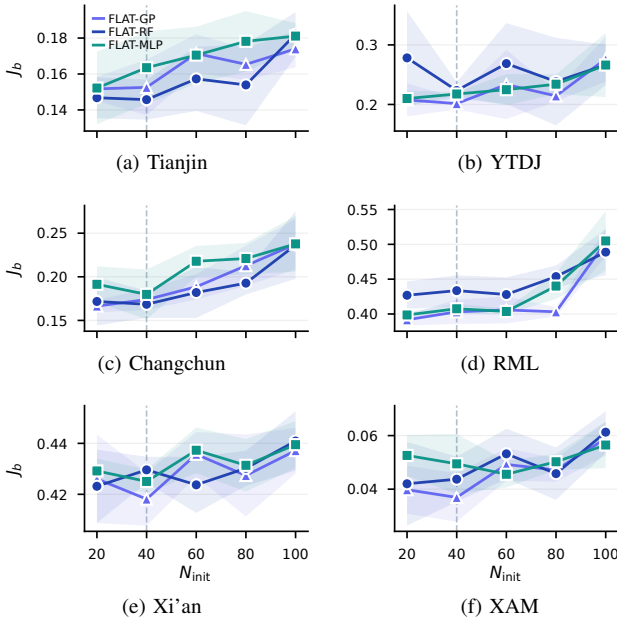


Fig. 7. N_{init} ablation: final J_b vs. warm-up size at fixed budget 100, five-seed mean ± 1 std. $N_{\text{init}}=100$ reduces to pure LHS.

to 0.28 at 100, 20% worse, with pure LHS worst or near-worst throughout (e.g. RML 0.403 $\rightarrow$ 0.503). Rugged scenes account for most of the penalty, whereas Xi'an shifts only 5% across the sweep, consistent with a fingerprint floor leaving little refinement headroom. $N_{\text{init}}=40$ matches the best average, cuts seed variance versus 20 (YTDJ std 0.010 vs. 0.028), and still leaves 60 LCB steps, matching the protocol above.

Surrogate choice is secondary. With J_b and acquisition fixed, GP, RF, and MLP track closely on average (Fig. 8). On easy scenes (XAM, Tianjin, Changchun; in-sample RF $R^2 \geq 0.97$) all three finish within 0.02. GP has the lowest global mean and tightest seed spread (Table I), fitting an uncertainty-calibrated LCB backend; on harder scenes (Xi'an, YTDJ, RML) MLP gains most, especially RML (J_b 0.434 $\rightarrow$ 0.408, R^2 0.976 $\rightarrow$ 0.997). No surrogate dominates; the GP-RF gap is at most 4.2%, versus 81% under objective ablation.

E. Operating Range

Sample-efficient guidance works where J_b is learnable yet baselines waste evaluations (YTDJ, RML, Changchun). XAM ($J_b \approx 0.044$) sets the easy bound, with FLAT, TPE, and GA within 0.01. Xi'an sets the hard bound at our SIND trio's high-saturation end, where signal cycling yields a stop-go fingerprint shared θ cannot match. Fig. 6 gives $\rho=0.56$ with speeds biased low relative to post-queue acceleration, a scene-wide instance of the fit trade-offs Punzo *et al.* [18] establish pairwise. Every optimizer family clusters at $J_b \approx 0.43$, SPSA included.

The bottleneck is Krauss car-following. Flat safe-speed logic misfits signal queues, while the Intelligent Driver Model (IDM) and psychophysical Wiedemann car-following laws encode stop-and-go transients instead. We keep Krauss and θ fixed under one TraCI protocol to isolate fingerprint smoothing and LCB

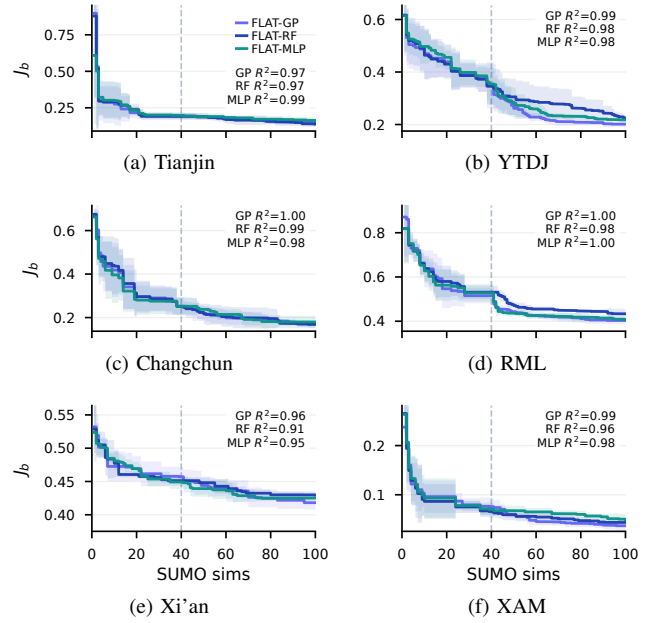


Fig. 8. Per-scene best-so-far J_b convergence for three surrogates, five-seed mean ± 1 std. R^2 annotations report surrogate fit at the final training set.

rather than car-following choice. Practical gains lie between XAM's basin and Xi'an's ceiling for scenes rugged enough to reward guidance yet below Xi'an's model limit.

VI. CONCLUSION

We presented FLAT, a framework that addresses the root cause of poor sample efficiency in surrogate-assisted traffic calibration: raw trajectory objectives are too rugged to be learned from few samples, leaving acquisition without a reliable signal. A behavioral fingerprint smooths the parameter-error landscape so that a sample surrogate can fit it from a few dozen samples, and annealed LCB acquisition then concentrates the remaining budget where it most reduces error.

Across six heterogeneous scenes and four optimizer families, FLAT is consistently best under matched budgets; even when allowed up to 500 simulations, competitors need 1.5 to 4.4 times more simulations to match FLAT's 100-run quality. Ablations confirm that the gain traces to objective geometry and sequential allocation rather than surrogate capacity: training on J_s instead of J_b raises final behavioral error by 81% on average, pure space-filling raises the six-scene mean by 20%, and the three surrogates differ by at most 4.2% in global mean. The finding suggests that objective geometry, not surrogate capacity, is the key bottleneck in sample-limited black-box optimization.

Future work will extend this insight through self-supervised encoders pretrained on corpora such as Waymo and nuPlan, deriving surrogate-learnable objectives offline and reserving scarce simulations for sequential refresh. Rolling digital twins at network scale should compose warm-started scene calibrations across observation windows, spending those budgets where mismatch drifts fastest. Transfer to other urban-mobility models and regional climate emulation remains to be tested.

APPENDIX A

SMOOTHNESS, LEARNABILITY AND UNCERTAINTY

This appendix traces whether slice-level smoothing persists in 10-D Θ , whether FLAT-GP can learn it at our calibration budget, and whether the resulting posterior supports (6). We report all three on XAM and Xi'an, our easy and hard bounds.

A. From Steepness to Smoothness

Fig. 9 extends Fig. 2 in the same $(a_{\max}, \tau)$ window, contrasting J_s and J_b with $|\nabla J|$. On XAM, J_b forms one wide, connected low-error basin with sparse gradient peaks, while J_s fragments into narrower, steeper cells. Xi'an is rougher for both objectives, yet J_b retains a single dominant basin where J_s does not. A sharp low- τ ridge marks queue discharge, where tighter headway triggers stop-and-go and a fingerprint jump.

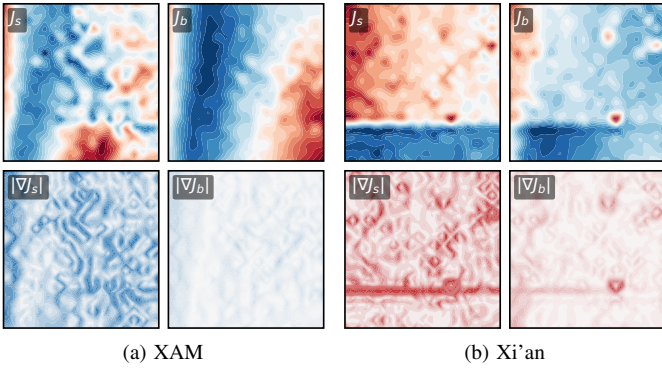


Fig. 9. Objective landscapes in the $(a_{\max}, \tau)$ plane on XAM and Xi'an. *Top row:* J_s and J_b . *Bottom row:* $|\nabla J_s|$ and $|\nabla J_b|$.

A 2-D slice cannot rule out this contrast as a projection artifact. We formalize the steepness above as a *local Lipschitz constant* on a 400-point LHS over 10-D θ . For each θ_i we take k nearest neighbors in the same min-max normalized $\tilde{\theta}$ as §IV-C and record the steepest slope,

$$\hat{L}_k(\theta_i) = \max_{j \in \mathcal{N}_k(\theta_i)} \frac{|J(\theta_i) - J(\theta_j)|}{\|\tilde{\theta}_i - \tilde{\theta}_j\|_2}, \quad (14)$$

with per-scene $\hat{L} = \text{p95}_i\{\hat{L}_k(\theta_i)\}$, robust to occasional SUMO replay noise. Fig. 10 plots $\hat{L}_{J_s}$ and $\hat{L}_{J_b}$ against k ; $\hat{L}_{J_s}$ exceeds $\hat{L}_{J_b}$ at every $k \in \{1, \dots, 10\}$ on both scenes, so the ranking is not a neighborhood-size artifact. At $k=5$, $\hat{L}_{J_s}/\hat{L}_{J_b}$ is $5.1\times$ on XAM and $3.1\times$ on Xi'an, narrowing on the harder scene.

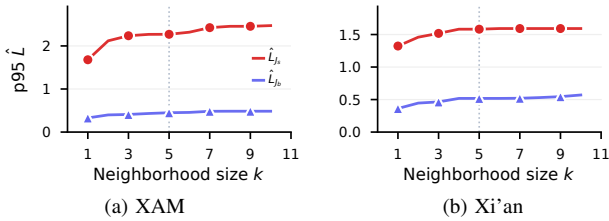


Fig. 10. Local Lipschitz estimate $\hat{L}$ (14) vs. neighborhood size k on a 10-D 400-point LHS design; dotted line marks the reported $k=5$.

B. From Smoothness to Learnability

Smoothness is necessary but not sufficient for a surrogate to learn the objective from few samples. On the same 10-D pool, we fit FLAT-GP on the first N points of a shared 300-point training pool and score a disjoint, fixed 100-point holdout set (Fig. 11), repeated over five subsamples. At $N=100$, held-out $R_{\text{ext}}^2(J_b)$ reaches 0.834 against 0.467 for J_s on XAM, and 0.634 against 0.453 on Xi'an, unlike the in-sample R^2 in Fig. 8, which tracks sequential calibration samples. At this budget J_b remains the learnable objective; we next test the posterior.

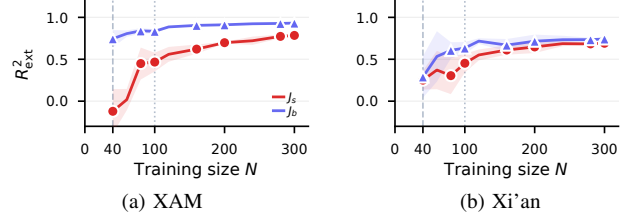


Fig. 11. Held-out R_{ext}^2 vs. training size N for FLAT-GP fit on J_s vs. J_b ; dotted line marks $N=100$, the calibration budget.

C. From Learnability to Uncertainty

On the same holdout split and $N=100$ budget as §A-B, we score the GP posterior (μ, σ) with negative log-likelihood (NLL) and continuous ranked probability score (CRPS), averaged over holdout points and five training subsamples:

$$\text{NLL} = \mathbb{E} \left[\frac{1}{2} \log(2\pi\sigma^2) + \frac{(y-\mu)^2}{2\sigma^2} \right], \quad (15)$$

$$\text{CRPS} = \mathbb{E} \int |F(z) - \mathbf{1}\{z \geq y\}| dz, \quad F = \Phi\left(\frac{\cdot - \mu}{\sigma}\right), \quad (16)$$

Table IV extends the $J_b \succ J_s$ ordering to probabilistic fit on both scenes: NLL stays negative and CRPS near zero for J_b with tight subsample spread, while J_s stays positive with CRPS elevated by 3–6 $\times$ and much wider NLL variance. Both scores test probabilistic consistency of the posterior with held-out data, asking whether σ tracks residual scale.

TABLE IV
GP PROBABILISTIC FIT ON HELD-OUT LANDSCAPES ($N=100$);
FIVE-SUBSAMPLE MEAN $\pm$ STD.

	XAM		Xi'an	
	J_b	J_s	J_b	J_s
NLL $\downarrow$	-1.563 ± 0.149	0.259 ± 0.087	-1.258 ± 0.173	0.093 ± 0.423
CRPS $\downarrow$	0.026 ± 0.002	0.169 ± 0.015	0.035 ± 0.003	0.106 ± 0.013

On J_s , that consistency breaks: the GP must spread uncertainty over a surface it cannot represent, so the σ -weighted exploration in (6) no longer points where error concentrates. Only J_b yields a well-calibrated posterior at our budget, closing Appendix A from landscape smoothness and held-out learnability to the uncertainty that (6) exploits.

APPENDIX B CALIBRATED BEHAVIORAL PARAMETERS

This appendix gives physical meanings and the search box Θ in Table V with a priori bounds that bracket plausible Krauss car-following and LC2013 lane-changing behavior across scenes, fixed before optimization for all methods.

TABLE V
TRACI KEYS, PHYSICAL MEANINGS, BOUNDS, AND UNITS FOR SIX
CAR-FOLLOWING AND FOUR LANE-CHANGING PARAMETERS.

TraCI key	Physical meaning	Bounds	Unit
<i>Car-following (Krauss)</i>			
accel	Maximum acceleration $a_{\max}$	[1.0, 3.5]	m/s ²
decel	Comfortable deceleration	[2.0, 6.0]	m/s ²
sigma	Driver imperfection	[0.1, 0.9]	—
tau	Desired time headway τ	[0.5, 2.5]	s
minGap	Minimum standing gap	[1.0, 4.0]	m
speedFactor	Desired-speed multiplier	[0.5, 1.2]	—
<i>Lane-changing (LC2013)</i>			
lcStrategic	Strategic/mandatory eagerness	[0.35, 2.80]	—
lcCooperative	Cooperative willingness	[0.35, 2.00]	—
lcAssertive	Lane-changing assertiveness	[0.35, 2.80]	—
lcSpeedGain	Discretionary speed-gain weight	[0.35, 2.80]	—

APPENDIX C OPTIMIZER PROTOCOLS AND BUDGET ACCOUNTING

All simulator evaluations are charged to the $B=100$ budget: FLAT’s 40-point LHS leaves 60 sequential LCB evaluations, while baseline initialization and update evaluations count toward the cap. Table VI reports the configurations used.

TABLE VI
OPTIMIZER PROTOCOLS AND EVALUATION-BUDGET ACCOUNTING FOR THE
MAIN COMPARISON.

Method	Protocol and budget
FLAT	40-point LHS + 60 sequential LCB evaluations
SPSA	Random start; $a=0.22$, $c=0.15$, $A=20$, $\alpha=0.602$, $\gamma=0.101$; evaluations capped at 100
GA	Random population of 10; tournament size 3; uniform crossover; mutation std 0.05 of span; population evaluations count toward cap
CMA-ES	Random start; normalized [0, 1] search; $\sigma_0=0.3$; implementation-default adaptation; evaluations capped at 100
TPE	Seeded Optuna TPESampler with implementation-default settings; 100 trials

REFERENCES

- [1] H. Xu, A. Berres, S. B. Yeginath, H. Sorensen, P. J. Nugent, J. Severino, S. A. Tennille, A. Moore, W. Jones, and J. Sanyal, “Smart mobility in the cloud: Enabling real-time situational awareness and cyber-physical control through a digital twin for traffic,” *IEEE Trans. Intell. Transp. Syst.*, vol. 24, no. 3, pp. 3145–3156, 2023.
- [2] P. A. Lopez, M. Behrisch, L. Bieker-Walz, J. Erdmann, Y.-P. Flötteröd, R. Hilbrich, L. Lücken, J. Rummel, P. Wagner, and E. Wießner, “Microscopic traffic simulation using SUMO,” in *Proc. 21st IEEE Int. Conf. Intelligent Transportation Systems (ITSC)*, 2018, pp. 2575–2582.
- [3] S. Krauß, “Microscopic modeling of traffic flow: Investigation of collision free vehicle dynamics,” Ph.D. dissertation, Universität zu Köln, 1998.
- [4] J. Erdmann, “Lane-changing model in SUMO,” in *Proc. SUMO 2014: Modeling Mobility with Open Data*, vol. 24. Berlin, Germany: DLR, 2014, pp. 77–88.
- [5] Y.-P. Flötteröd and M. Behrisch, “Online calibration with SUMO for network-wide traffic and emission monitoring: Case study ITS huainan,” in *Proc. SUMO User Conference*, ser. SUMO Conference Proceedings, vol. 2. TIB Open Publishing, 2022, pp. 1–11.
- [6] D. E. Goldberg, *Genetic Algorithms in Search, Optimization, and Machine Learning*. Addison-Wesley, 1989.
- [7] J. C. Spall, “Multivariate stochastic approximation using a simultaneous perturbation gradient approximation,” *IEEE Trans. Autom. Control*, vol. 37, no. 3, pp. 332–341, 1992.
- [8] N. Hansen and A. Ostermeier, “Completely derandomized self-adaptation in evolution strategies,” *Evolutionary Computation*, vol. 9, no. 2, pp. 159–195, 2001.
- [9] D. R. Jones, M. Schonlau, and W. J. Welch, “Efficient global optimization of expensive black-box functions,” *Journal of Global Optimization*, vol. 13, no. 4, pp. 455–492, 1998.
- [10] B. Shahriari, K. Swersky, Z. Wang, R. P. Adams, and N. de Freitas, “Taking the human out of the loop: A review of Bayesian optimization,” *Proc. IEEE*, vol. 104, no. 1, pp. 148–175, 2016.
- [11] J. Bergstra, R. Bardenet, Y. Bengio, and B. Kégl, “Algorithms for hyperparameter optimization,” in *Advances in Neural Information Processing Systems (NeurIPS)*, 2011, pp. 2546–2554.
- [12] J. Snoek, H. Larochelle, and R. P. Adams, “Practical Bayesian optimization of machine learning algorithms,” in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 25, 2012, pp. 2951–2959.
- [13] F. Hutter, H. H. Hoos, and K. Leyton-Brown, “Sequential model-based optimization for general algorithm configuration,” in *Proc. Learning and Intelligent Optimization (LION)*, ser. LNCS, vol. 6683, 2011, pp. 507–523.
- [14] S. Agriesti, V. Kuzmanovski, J. Hollmén, C. Roncoli, and B. Nahmias-Biran, “A Bayesian optimization approach for calibrating large-scale activity-based transport models,” *IEEE Open J. Intell. Transp. Syst.*, vol. 4, pp. 740–754, 2023.
- [15] P. Mangano-Redondo, A. Paricio-Garcia, and M. A. Lopez-Carmona, “Spatio-temporal AI modeling for urban traffic calibration: A SUMO-based approach,” in *Proc. SUMO User Conference*, ser. SUMO Conference Proceedings, vol. 6. TIB Open Publishing, 2025, pp. 33–50.
- [16] S. Ryu, D. Kwon, S. Choi, A. Deshwal, S. Kang, and C. Osorio, “BO4Mob: Bayesian optimization benchmarks for high-dimensional urban mobility problem,” in *Advances in Neural Information Processing Systems (NeurIPS)*, 2025, datasets and Benchmarks Track.
- [17] B. Ciuffo and V. Punzo, ““no free lunch” theorems applied to the calibration of traffic simulation models,” *IEEE Trans. Intell. Transp. Syst.*, vol. 15, no. 2, pp. 553–562, 2014.
- [18] V. Punzo, Z. Zheng, and M. Montanino, “About calibration of car-following dynamics of automated and human-driven vehicles: Methodology, guidelines and codes,” *Transportation Research Part C: Emerging Technologies*, vol. 128, p. 103165, 2021.
- [19] M. Schrader, A. Karnik, A. Hainen, and J. Bittle, “Calibrating car-following models using SUMO-in-the-loop and vehicle trajectories from roadside radar: Calibrating CF model parameters,” in *Proc. SUMO User Conference*, ser. SUMO Conference Proceedings, vol. 5. TIB Open Publishing, 2024, pp. 209–233.
- [20] I. M. A. Abuamer and C. M. J. Tampère, “Traffic microsimulation for smart cities: Investigating the impact of objective function formulation on calibration efficiency,” *IET Smart Cities*, vol. 6, no. 4, pp. 276–290, 2024.
- [21] R. Moriconi, M. P. Deisenroth, and K. S. Sesh Kumar, “High-dimensional Bayesian optimization using low-dimensional feature spaces,” *Machine Learning*, vol. 109, no. 9, pp. 1925–1943, 2020.
- [22] A. Zharmagambetov, B. Amos, A. Ferber, T. Huang, B. Dilkina, and Y. Tian, “Landscape surrogate: Learning decision losses for mathematical optimization under partial information,” in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 36, 2023, pp. 27 332–27 350.
- [23] D. Eriksson, M. Pearce, J. Gardner, R. D. Turner, and M. Poloczek, “Scalable global optimization via local Bayesian optimization,” in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 32, 2019, pp. 5496–5507.
- [24] Federal Highway Administration, “Definition, interpretation, and calculation of traffic analysis tools measures of effectiveness,” U.S. Department of Transportation, FHWA, Tech. Rep. HOP-08-054, 2008, traffic Analysis Toolbox Vol. VI. [Online]. Available: <https://ops.fhwa.dot.gov/publications/fhwahop08054/>
- [25] Transportation Research Board, *Highway Capacity Manual*, 6th ed. Washington, DC: Transportation Research Board, 2016.

- [26] M. D. McKay, R. J. Beckman, and W. J. Conover, "A comparison of three methods for selecting values of input variables in the analysis of output from a computer code," *Technometrics*, vol. 21, no. 2, pp. 239–245, 1979.
- [27] B. Lakshminarayanan, A. Pritzel, and C. Blundell, "Simple and scalable predictive uncertainty estimation using deep ensembles," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2017, pp. 6402–6413.
- [28] Y. Xu, W. Shao, J. Li, K. Yang, W. Wang, H. Huang, C. Lv, and H. Wang, "SIND: A drone dataset at signalized intersection in china," in *Proc. 25th IEEE Int. Conf. Intelligent Transportation Systems (ITSC)*, 2022, pp. 2471–2478.
- [29] R. Feng, H.-X. Zhu, N. N. Sze, S. Wang, and Z. Li, "Ubiquitous traffic eyes: Trajectory dataset focus on multiple traffic states and state transition on urban expressways," *Transportation Letters*, vol. 18, no. 2, pp. 446–462, 2025.
- [30] F. Wilcoxon, "Individual comparisons by ranking methods," *Biometrics Bulletin*, vol. 1, no. 6, pp. 80–83, 1945.